4.1 p.195 # 1, 3, 5, 8, 11, 13, 15, 17, 35

1. $V = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x \geq 0, y \geq 0 \right\}$

a) $u = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$ $v = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$ $u+v = \begin{bmatrix} x_1+x_2 \\ y_1+y_2 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$ yes

b) $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $c = -2$ $cu = \begin{bmatrix} -2 \\ -4 \end{bmatrix}$ not in V

3. $H = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x^2 + y^2 \leq 1 \right\}$

$u = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $u+v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ not in H

5. $p(t) = at^2$
set is the span$\{t^2\}$ so subspace by Thm 1

8. P_n such that $p(0) = 0$
a) ~~zero~~ in
b) $(p+q)(0) = p(0) + q(0) = 0+0 = 0$
c) $(cp)(0) = c\,p(0) = c \cdot 0 = 0$

11. $\begin{bmatrix} 2b+3c \\ -b \\ 2c \end{bmatrix}$ $b\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + c\begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$ $W = \text{span}\left\{ \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} \right\}$

Subspace $\mathbb{R}^3$ by Thm 1

13. a) no, just those 3
b) ∞
c) $\begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 1 & 2 & 1 \\ -1 & 3 & 6 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ so yes

15. $\begin{bmatrix} 2a+3b \\ -1 \\ 2a-5b \end{bmatrix}$ 0 vector not in it ∴ not subspace

17. $\begin{bmatrix} 2a-b \\ 3b-c \\ 3c-a \\ 3b \end{bmatrix} = a\begin{bmatrix} 2 \\ 0 \\ -1 \\ 0 \end{bmatrix} + b\begin{bmatrix} -1 \\ 3 \\ 0 \\ 3 \end{bmatrix} + c\begin{bmatrix} 0 \\ -1 \\ 3 \\ 0 \end{bmatrix}$

35. $\begin{bmatrix} 8 & -4 & -7 & 9 \\ -4 & 3 & 6 & -4 \\ -3 & -2 & -5 & -4 \\ 9 & -8 & -18 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ w in subspace (consistent matrix)

4.2 p. 205 #1, 5, 9, 13, 15, 17*, 19*, 23, 25, 31, 34, 37

1. $Aw = \begin{bmatrix} 3 & -5 & -3 \\ 6 & -2 & 0 \\ -8 & 4 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 3 \\ -4 \end{bmatrix} = \begin{bmatrix} 3-15+12 \\ 6-6+0 \\ -8+12-4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ yes

5. $\begin{bmatrix} 1 & -4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -5 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$ $x_1 = 4x_2 - 2x_4$, $x_3 = 5x_4$, $2x_5 = 0$

$x_2\begin{bmatrix} 4 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4\begin{bmatrix} -2 \\ 0 \\ 5 \\ 1 \\ 0 \end{bmatrix}$

9. $\left\{ \begin{bmatrix} p \\ q \\ r \\ s \end{bmatrix} : \begin{matrix} p-3q=4s \\ 2p=s+5r \end{matrix} \right\}$ $A = \begin{bmatrix} 1 & -3 & 0 & -4 \\ 2 & 0 & -5 & -1 \end{bmatrix}$

$W = \text{Nul } A$ so subspace $\mathbb{R}^4$ by thm 2 + is a vector space

13. $\left\{ \begin{bmatrix} c-6d \\ d \\ c \end{bmatrix} : c, d \in \mathbb{R} \right\} = c\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + d\begin{bmatrix} -6 \\ 1 \\ 0 \end{bmatrix}$ so

$= \text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -6 \\ 1 \\ 0 \end{bmatrix} \right\}$ and is a subspace of $\mathbb{R}^3$ by Thm 3 + vector space

15. $\begin{bmatrix} 2s+t \\ r-s+2t \\ 3r+s \\ 2r-s-t \end{bmatrix}$ $s\begin{bmatrix} 2 \\ -1 \\ 1 \\ -1 \end{bmatrix} + t\begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix} + r\begin{bmatrix} 0 \\ 1 \\ 3 \\ 2 \end{bmatrix}$ $A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 0 \\ 2 & -1 & -1 \end{bmatrix}$

17. $A=\begin{bmatrix}6 & -4\\ -3 & 2\\ -9 & 6\\ 9 & -6\end{bmatrix}$ 4×2

a) Nul A subspace $\mathbb{R}^2$
b) Col A subspace $\mathbb{R}^4$

19. $A=\begin{bmatrix}4 & 5 & -2 & 6 & 0\\ 1 & 1 & 0 & 1 & 0\end{bmatrix}$ 2×5

a) Nul A subspace $\mathbb{R}^5$
b) Col A " $\mathbb{R}^2$

23. $A=\begin{bmatrix}-2 & 4\\ -1 & 2\end{bmatrix}$ $w=\begin{bmatrix}2\\ 1\end{bmatrix}$ $Aw=\begin{bmatrix}0\\ 0\end{bmatrix}$ so $w\in$ Nul A

$\begin{bmatrix}-2 & 4 & 2\\ -1 & 2 & 1\end{bmatrix}\sim\begin{bmatrix}1 & -2 & -1\\ -2 & 4 & 2\end{bmatrix}\sim\begin{bmatrix}1 & -2 & -1\\ 0 & 0 & 0\end{bmatrix}$ so $w\in$ Col A

25. a) T
b) F ($\mathbb{R}^n$)
c) T
d) F ($\neq b$?)
e) T (Nul)
f) T

31. a) $p=a_1+b_1t+c_1t^2$
$q=a_2+b_2t+c_2t^2$

$T(p)=\begin{bmatrix}a_1\\ a_1+b_1+c_1\end{bmatrix}$ $T(q)=\begin{bmatrix}a_2\\ a_2+b_2+c_2\end{bmatrix}$

$T(p+q)=\begin{bmatrix}(p+q)(0)\\ (p+q)(1)\end{bmatrix}=\begin{bmatrix}p(0)+q(0)\\ p(1)+q(1)\end{bmatrix}=\begin{bmatrix}p(0)\\ p(1)\end{bmatrix}+\begin{bmatrix}q(0)\\ q(1)\end{bmatrix}=T(p)+T(q)$

$T(cp)=\begin{bmatrix}(cp)(0)\\ (cp)(1)\end{bmatrix}=c\begin{bmatrix}p(0)\\ p(1)\end{bmatrix}=cT(p)$

b) a) if $q(0)$ and $q(1)=0$, q in Kernel of T
so $q(t)=cp(t)$ where $p=t(t-1)$. b) for any vector $\begin{bmatrix}x_2\\ x_2\end{bmatrix}$ in $\mathbb{R}^2$
$p=x_1+(x_2-x_1)$ has $p(0)=x_1$ $p(1)=x_2$. Thus Range T is $\mathbb{R}^2$

34. f, g in $C[0,1]$, c scalar

$F = T(f) = \int f$, $F(0)=0$ $\quad G = T(g) = \int g$, $G(0)=0$

$\int(f+g) = F+G$, $(F+G)(0) = F(0)+G(0) = 0$

$T(f+g) = T(f)+T(g)$

$\int cf = cF$, $(cF)(0) = cF(0) = 0$

$T(cf) = cT(f)$

} T linear trans.

all functions f w/ $\int f = 0$ → kernel

only one is 0

so kernel T is $\{0\}$

37. $A = \begin{bmatrix} 7 & 6 & -4 & 0 \\ -5 & -1 & 0 & -2 \\ 9 & -11 & 7 & -3 \\ 19 & -9 & 7 & 0 \end{bmatrix}$ $\quad w = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}$

$[A\ w]$ consistent

so $w \in$ col A

$Aw = \begin{bmatrix} 14 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ so w not in nul A.

4.3 p. 213 # 1-13 odd, 17, 19, 21, 33, 36

1. $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}$

yes (pivots)

3. $\begin{bmatrix}1\\0\\-3\end{bmatrix}, \begin{bmatrix}3\\1\\-4\end{bmatrix}, \begin{bmatrix}-2\\-1\\1\end{bmatrix} \sim \begin{bmatrix}1&3&-2\\0&1&-1\\0&5&-5\end{bmatrix} \sim \begin{bmatrix}1&3&-2\\0&1&-1\\0&0&0\end{bmatrix}$

lin. dep, doesn't span $\mathbb{R}^3$

5. $\begin{bmatrix}3\\-3\\0\end{bmatrix}, \begin{bmatrix}-3\\7\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\-3\\5\end{bmatrix} \sim \begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\end{bmatrix}$

not lin. indep. (0)

piv. each so does span.

7. $\begin{bmatrix}-2\\3\\0\end{bmatrix}, \begin{bmatrix}6\\-1\\5\end{bmatrix}$ can't have pivot each row so doesn't span

$\sim \begin{bmatrix}1&0\\0&1\\0&0\end{bmatrix}$ is lin. indep.

9. $\begin{bmatrix}1&0&-2&-2\\0&1&1&4\\3&-1&-7&3\end{bmatrix} \sim \begin{bmatrix}1&0&-2&-2\\0&1&1&4\\0&-1&-1&9\end{bmatrix} \sim \begin{bmatrix}1&0&-2&-2\\0&1&1&4\\0&0&0&13\end{bmatrix} \sim \begin{bmatrix}1&0&-2&-2\\0&1&1&4\\0&0&0&1\end{bmatrix}$

$\sim \begin{bmatrix}1&0&-2&0\\0&1&1&0\\0&0&0&1\end{bmatrix}$ $x_1 = 2x_3$, $x_2 = -x_3$, $x_4 = 0$

$x_3 \begin{bmatrix}2\\-1\\1\\0\end{bmatrix}$ basis Nul $A = \left\{\begin{bmatrix}2\\-1\\1\\0\end{bmatrix}\right\}$

11. $x - 3y + 2z = 0$

$\begin{bmatrix}1&-3&2\end{bmatrix}$ $x_1 = 3x_2 - 2x_3$

$x_2 \begin{bmatrix}3\\1\\0\end{bmatrix} + x_3 \begin{bmatrix}-2\\0\\1\end{bmatrix}$ basis: $\left\{\begin{bmatrix}3\\1\\0\end{bmatrix}, \begin{bmatrix}-2\\0\\1\end{bmatrix}\right\}$

13. $\begin{bmatrix}-2&4&-2&-4\\2&-6&-3&1\\-3&8&2&-3\end{bmatrix} \sim \begin{bmatrix}1&0&6&5\\0&2&5&3\\0&0&0&0\end{bmatrix}$ basis Col $A = \left\{\begin{bmatrix}-2\\2\\-3\end{bmatrix}, \begin{bmatrix}4\\-6\\8\end{bmatrix}\right\}$

$x_1 = -6x_3 - 5x_4$

$x_2 = -\frac{5}{2}x_3 - \frac{3}{2}x_4$

$x_3 \begin{bmatrix}-6\\-5/2\\1\\0\end{bmatrix} + x_4 \begin{bmatrix}-5\\-3/2\\0\\1\end{bmatrix}$ basis Nul $A = \left\{\begin{bmatrix}-6\\-5/2\\1\\0\end{bmatrix}, \begin{bmatrix}-5\\-3/2\\0\\1\end{bmatrix}\right\}$

17. $\begin{bmatrix}2&4&-2&8&-8\\0&0&-4&4&4\\-4&2&0&8&0\\-6&-4&1&-3&0\\0&4&7&15&1\end{bmatrix} \sim \begin{bmatrix}1&0&0&-1&0\\0&1&0&2&0\\0&0&1&-1&0\\0&0&0&0&1\\0&0&0&0&0\end{bmatrix}$

basis: $\left\{\begin{bmatrix}2\\0\\-4\\-6\\0\end{bmatrix}, \begin{bmatrix}4\\0\\2\\-4\\4\end{bmatrix}, \begin{bmatrix}-2\\-4\\0\\1\\7\end{bmatrix}, \begin{bmatrix}-8\\4\\0\\0\\1\end{bmatrix}\right\}$

19. - Since $4v_1 + 5v_2 - 3v_3 = 0$, any v can be written as linear combo of other 2
- none multiple of others
so basis any 2/3

21. a. False (only zero vector)
b. False (and $\{b_1, \dots b_p\}$ lin indep.)
c. True
d. False (spanning set as small as possible, lin ind set as large as possible)
e. False

33. $p_1 = 1 + t^2$, $p_2 = 1 - t^2$ $\quad p_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ $p_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$

not multiples so yes, lin. ind. set (in P_3 and P_2)

36. $\begin{bmatrix} 1 & 0 & 3 \\ 2 & 2 & 4 \\ 0 & -1 & 1 \\ -1 & 1 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

basis H: $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -1 \\ 1 \end{bmatrix} \right\}$
$u_1 \quad u_2$

$\begin{bmatrix} -2 & 2 & -1 \\ -2 & 3 & 4 \\ -1 & 2 & 6 \\ 3 & -6 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

basis K: $\{v_1, v_2, v_3\}$

$[u_1\ u_2\ u_3\ v_1\ v_2\ v_3] \sim \begin{bmatrix} 1 & 0 & 3 & -2 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

basis H+K:
$\{u_1, u_2, v_2, v_3\}$

4.4 p. 222 # 3, 7, 9, 11, 13, 17, 21, 27, 29, 31, 33

$x = P_B [x]_B$

3. $\begin{bmatrix} 1 & 5 & 4 \\ -2 & 0 & -3 \\ 3 & -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 1+0-8 \\ -2+0+6 \\ 3+0+0 \end{bmatrix} = \begin{bmatrix} -7 \\ 4 \\ 3 \end{bmatrix} = \bar{x}$

7. $\begin{bmatrix} 1 & -3 & 2 & 8 \\ -1 & 4 & -2 & -9 \\ -3 & 9 & 4 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 10 & 30 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix}$

$\uparrow$ $[x]_B$

9. $\beta = \left\{ \begin{bmatrix} 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 2 \\ -5 \end{bmatrix} \right\}$ then $P_B = \begin{bmatrix} 1 & 2 \\ -3 & -5 \end{bmatrix}$

11. $P_B = \begin{bmatrix} 1 & -3 \\ -2 & 5 \end{bmatrix}$ $\quad P_B^{-1} = \frac{1}{5-6} \begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -5 & -3 \\ -2 & -1 \end{bmatrix}$

$[x]_B = \begin{bmatrix} -5 & -3 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -5 \end{bmatrix} = \begin{bmatrix} -10+15 \\ -4+5 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$

13. $[b_1\ b_2\ b_3\ p] \sim [e_1\ e_2\ e_3\ [p]_B]$

$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 1 & 1 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & 0 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & -2 & 2 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & -1 \end{bmatrix}$ $\quad [p]_B = \begin{bmatrix} 2 \\ 6 \\ -1 \end{bmatrix}$

17. $x_1 \begin{bmatrix} 1 \\ -3 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -8 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 2 & -3 & 1 \\ -3 & -8 & 7 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 & 1 \\ 0 & -2 & -2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -5 & 5 \\ 0 & 1 & 1 & -2 \end{bmatrix}$

$x_1 = 5 + 5x_3$

$x_2 = -2 - x_3$

$x_3 = 0$: $5v_1 - 2v_2$

$x_3 = 1$: $10v_1 - 3v_2 + v_3$ } 2 ex.

21. $\begin{bmatrix} 1 & -2 \\ -4 & 9 \end{bmatrix}^{-1} = \frac{1}{9-8}\begin{bmatrix} 9 & 2 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 9 & 2 \\ 4 & 1 \end{bmatrix}$

27. using standard basis for P_3,

$[x_1]_B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix}$ $[x_2]_B = \begin{bmatrix} 2 \\ 1 \\ -3 \\ 0 \end{bmatrix}$ $[x_3]_B = \begin{bmatrix} 0 \\ -1 \\ 2 \\ -1 \end{bmatrix}$

$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & -3 & 2 \\ 2 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

pivot each column, so columns (and polynomials) lin. indep.

29. same basis as #27 $1-2t+t^2$, $t-2t^2+t^3$, $1-3t+3t^2-t^3$

$\begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & -3 \\ 1 & -2 & 3 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

not a pivot each column

$\therefore$ not lin. indep.

31. a) $\begin{bmatrix} 1 & -3 & -4 & 1 \\ -3 & 5 & 5 & 0 \\ 5 & -7 & -6 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -4 & 1 \\ 0 & -4 & -7 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

not a pivot each row $\therefore$ don't span $\mathbb{R}^3$. $\mathbb{R}^3$ isomorphic to P_2 so don't span P_2

b) $\begin{bmatrix} 0 & 1 & -3 & 2 \\ 5 & -8 & 4 & -3 \\ 1 & -2 & 2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 2 & 0 \\ 0 & 2 & -6 & -3 \\ 0 & 0 & 0 & 7/2 \end{bmatrix}$

does span $\mathbb{R}^3$ b/c pivot each row $\therefore$ spans P_2

33. $\begin{bmatrix} 3 & 5 & 0 & 1 \\ 7 & 1 & 1 & 16 \\ 0 & 0 & -2 & -6 \\ 0 & -2 & 0 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

not a basis for $\mathbb{R}^4$
- not lin. indep (piv. in col)
- not spanning (piv in row)

so not a basis for P_3

4.5 p.229 #1, 5, 7, 11, 13, 15, 17, 21

1. $\begin{bmatrix} s-2t \\ s+t \\ 3t \end{bmatrix} = s\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix}$ basis: $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix} \right\}$ dim = 2

5. $\begin{bmatrix} p-2q \\ 2p-5r \\ -2q+2r \\ -3p+6r \end{bmatrix} = p\begin{bmatrix} 1 \\ 2 \\ 0 \\ -3 \end{bmatrix} + q\begin{bmatrix} -2 \\ 0 \\ -2 \\ 0 \end{bmatrix} + r\begin{bmatrix} 0 \\ -5 \\ 2 \\ 6 \end{bmatrix}$

row reduce → pivot each col.

∴ basis and dim = 3

7. $a\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b\begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} + c\begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 3 & 0 \end{bmatrix} \sim I$

$A = \begin{bmatrix} 1 & -3 & 1 \\ 0 & 1 & -2 \\ 0 & 2 & -1 \end{bmatrix}$ subspace $H = \text{Nul } A$ only trivial so $H = \text{Nul } A = \{0\}$, $\dim H = 0$

11. $\begin{bmatrix} 1 & 3 & -2 & 5 \\ 0 & 1 & -1 & 2 \\ 2 & 1 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\dim \text{Col } A = 3$ (3 pivots)

13. $A = \begin{bmatrix} 1 & -6 & 9 & 0 & -2 \\ 0 & 1 & 2 & -4 & 5 \\ 0 & 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ 3 pivots ∴ $\dim \text{Col } A = 3$; 2 free ∴ $\dim \text{Nul } A = 2$

15. $A = \begin{bmatrix} 1 & 2 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$ 3 pivots, 2 free; $\dim \text{Col } A = 3$, $\dim \text{Nul } A = 2$

17. $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$ 3 pivots, no free; $\dim \text{Col } A = 3$, $\dim \text{Nul } A = 0$

21. $\begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 2 & 0 & -12 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}$

$\mathbb{R}^4$ isomorphic to $\mathbb{P}_3$

pivots each col. $\therefore$ lin. indep. 4 poly and $\dim \mathbb{P}_3 = 4$ $\therefore$ Basis Thm says basis for $\mathbb{P}_3$

46 p:236 # 3, 5-16, 17, 27, 35

3. $A=\begin{bmatrix}2&6&-6&6&3&6\\-2&-3&6&-3&0&-6\\4&9&-12&9&3&12\\-2&3&6&3&3&-6\end{bmatrix} \sim B=\begin{bmatrix}2&6&-6&6&3&6\\0&3&0&3&3&0\\0&0&0&0&3&0\\0&0&0&0&0&0\end{bmatrix}$

$\dim \text{Col } A = \text{rank } A = 3$

$\dim \text{Nul } A = 3$

basis Col $A = \left\{\begin{bmatrix}2\\-2\\4\\-2\end{bmatrix}, \begin{bmatrix}6\\-3\\9\\3\end{bmatrix}, \begin{bmatrix}3\\0\\3\\3\end{bmatrix}\right\}$ basis Row $A = \left\{\begin{bmatrix}2&6&-6&6&3&6\end{bmatrix}, \begin{bmatrix}0&3&0&3&3&0\end{bmatrix}, \begin{bmatrix}0&0&0&0&3&0\end{bmatrix}\right\}$

$A \sim \begin{bmatrix}1&0&-3&0&0&3\\0&1&0&1&0&0\\0&0&0&0&1&0\\0&0&0&0&0&0\end{bmatrix}$ $x_1 = 3x_3 - 3x_6$, $x_2 = -x_4$, $x_5 = 0$

basis Nul $A = \left\{\begin{bmatrix}3\\0\\1\\0\\0\\0\end{bmatrix}\begin{bmatrix}0\\-1\\0\\1\\0\\0\end{bmatrix}\begin{bmatrix}-3\\0\\0\\0\\0\\1\end{bmatrix}\right\}$

5. 4×7 rank $= 3$, $\dim \text{Nul} = 4$, $\dim \text{Row} = 3$, rank $A^T = 3$

6. 7×5 rank $= 2$, $\dim \text{Nul} = 3$, $\dim \text{Row} = \text{rank } A^T = 2$

7. 4×7, 4 pivots $\text{Col } A = \mathbb{R}^4$ because $\dim \text{col } A = 4$ and 4 rows

$\dim \text{Nul } A = 3$ but subspace of $\mathbb{R}^7$ so $\text{Nul } A \neq \mathbb{R}^3$

8. 6×8, 4 pivots $\dim \text{Nul } A = 4$ $(8-4=4)$

$\dim \text{Col } A = 4$ but in 6 dimensions so $\text{Col } A \neq \mathbb{R}^4$ (so $\mathbb{R}^4$ subspace)

9. 4×6, $\dim \text{Nul } A = 3$ $\dim \text{Col } A = 6-3 = 3$

but Col A subspace $\mathbb{R}^4$ so $\neq \mathbb{R}^3$

10. 8×7, $\dim \text{Nul } A = 5$ $\dim \text{Col } A = 7-5 = 2$

11. 8x5, $\dim \text{Nul } A = 3$ $\dim \text{row } A = \text{rank } A = 5-3 = 2$

12. 5x4, $\dim \text{Nul } A = 2$ $\dim \text{row } A = \text{rank } A = 4-2 = 2$

13. 7x5 largest rank $A = 5$ (can have pivot each col)
5x7 largest rank $A = 5$ (most pivots possible)

14. 5x4 largest $\dim \text{row } A = 4$ same as rank
4x5

15. 3x7 Smallest $\dim \text{Nul } A = 4$ most 3 pivots so 4 free

16. 7x5 smallest $\dim \text{Nul } A = 0$ can have all 5 pivots

17. a) T
b) F (rows of B)
c) T (Rank Thm)
d) F (columns)
e) T (numerical note)

27. $\mathbb{R}^m$: Col A, Row A^T, Nul A^T $\mathbb{R}^n$: Row A, Nul A, Col A^T
4 distinct. (col A = row A^T, row A = col A^T)

35. a) $A \sim \begin{bmatrix} 1 & 0 & 13/2 & 0 & 5 & 0 & -3 \\ 0 & 1 & 11/2 & 0 & 1/2 & 0 & 2 \\ 0 & 0 & 0 & 1 & -11/2 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ ↑ ↑ ↑ ↑

$C = \begin{bmatrix} 7 & -9 & 5 & -3 \\ -4 & 6 & -2 & -5 \\ 5 & -7 & 5 & 2 \\ -3 & 5 & -1 & -4 \\ 6 & -8 & 4 & 9 \end{bmatrix}$

$R = \begin{bmatrix} 1 & 0 & 13/2 & 0 & 5 & 0 & -3 \\ 0 & 1 & 11/2 & 0 & 1/2 & 0 & 2 \\ 0 & 0 & 0 & 1 & -11/2 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$

$N = \begin{bmatrix} -13/2 & -5 & 3 \\ -11/2 & -1/2 & -2 \\ 1 & 0 & 0 \\ 0 & 11/2 & -7 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

b) $A^T \sim \begin{bmatrix} 1 & 0 & 0 & 0 & -2/11 \\ 0 & 1 & 0 & 0 & -41/11 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 28/11 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$M = \begin{bmatrix} 2/11 \\ 41/11 \\ 0 \\ -28/11 \\ 1 \end{bmatrix}$

$S = [R^T N]$ is 7x7
b/c R^T and N in $\mathbb{R}^7$ and
rank A^T + dim Nul $A^T = 7$

$T = [C \; M]$ is 5x5 invertible b/c col. lin. indep

4.7 p.242 # 1, 3, 5, 9, 13, 17, 18

1. $b_1 = 6c_1 - 2c_2 \quad b_2 = 9c_1 - 4c_2$

a) $[b_1]_C = \begin{bmatrix} 6 \\ -2 \end{bmatrix} \quad [b_2]_C = \begin{bmatrix} 9 \\ -4 \end{bmatrix} \quad \underset{C \leftarrow B}{P} = \begin{bmatrix} 6 & 9 \\ -2 & -4 \end{bmatrix}$

b) $\begin{bmatrix} 6 & 9 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} -3 \\ 2 \end{bmatrix} = \underset{C \leftarrow B}{P} [x]_B = [x]_C = \begin{bmatrix} -18+18 \\ 6-8 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$

3. $P = \begin{bmatrix} [u_1]_w & [u_2]_w \end{bmatrix} = \underset{w \leftarrow u}{P}$ ii) $[x]_w = P[x]_u$

5. $a_1 = 4b_1 - b_2 \quad a_2 = -b_1 + b_2 + b_3 \quad a_3 = b_2 - 2b_3$

a) $\underset{B \leftarrow A}{P} = \begin{bmatrix} [a_1]_B & [a_2]_B & [a_3]_B \end{bmatrix} = \begin{bmatrix} 4 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & -2 \end{bmatrix}$

b) $\begin{bmatrix} 4 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 12-4+0 \\ -3+4+1 \\ 0+4-2 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \\ 2 \end{bmatrix} = [x]_B$

9. $\underset{C \leftarrow B}{P}$, $\begin{bmatrix} 2 & -2 & 4 & 8 \\ 2 & 2 & 4 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 & 4 \\ 1 & 1 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 & 4 \\ 0 & 2 & 0 & -2 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & -1 & 2 & 4 \\ 0 & 1 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & -1 \end{bmatrix} \quad \underset{C \leftarrow B}{P} = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix} \quad \underset{B \leftarrow C}{P} = \frac{-1}{2} \begin{bmatrix} -1 & -3 \\ 0 & 2 \end{bmatrix}$

13. $[b_1]_C = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \quad [b_2]_C = \begin{bmatrix} 3 \\ -5 \\ 4 \end{bmatrix} \quad [b_3]_C = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} \quad [x]_C = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$

$\underset{C \leftarrow B}{P} = \begin{bmatrix} 1 & 3 & 0 \\ -2 & -5 & 2 \\ 1 & 4 & 3 \end{bmatrix} \quad [x]_B = \underset{C \leftarrow B}{P^{-1}} [x]_C$

$\begin{bmatrix} 1 & 3 & 0 & -1 \\ -2 & -5 & 2 & 2 \\ 1 & 4 & 3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad [x]_B = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$

$\cos 2t = -1 + 2\cos^2 t$

$\cos 3t = -3\cos t + 4\cos^3 t$

$\cos 4t = 1 - 8\cos^2 t + 8\cos^4 t$

$\cos 5t = 5\cos t - 20\cos^3 t + 16\cos^5 t$

$\cos 6t = -1 + 18\cos^2 t - 48\cos^4 t + 32\cos^6 t$

$H = \text{Span}\{x_0 \dots x_6\}$

17. $B = \{x_0 \dots x_6\}$, $x_k = \cos^k t$, $C = \{y_0 \dots y_6\}$, $y_k = \cos kt$

a)

$$P = \begin{bmatrix} [y_0]_B & \cdots & [y_6]_B \end{bmatrix} = \underset{B \leftarrow C}{P}$$

$$P = \begin{bmatrix} 1 & 0 & -1 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & -3 & 0 & 5 & 0 \\ 0 & 0 & 2 & 0 & -8 & 0 & 18 \\ 0 & 0 & 0 & 4 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & 8 & 0 & -48 \\ 0 & 0 & 0 & 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 32 \end{bmatrix}$$

$$P^{-1} = \frac{1}{32}\begin{bmatrix} 32 & 0 & 16 & 0 & 12 & 0 & 10 \\ 0 & 32 & 0 & 24 & 0 & 20 & 0 \\ 0 & 0 & 16 & 0 & 16 & 0 & 15 \\ 0 & 0 & 0 & 8 & 0 & 10 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 & 6 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} \quad \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \end{matrix}$$

$$= \underset{C \leftarrow B}{P}$$

$\cos^2 t = \frac{1}{2} + \frac{1}{2}\cos 2t$

$\cos^3 t = \frac{3}{4}\cos t + \frac{1}{4}\cos 3t$

$\cos^4 t = \frac{3}{8} + \frac{1}{2}\cos 2t + \frac{1}{8}\cos 4t$

$\cos^5 t = \frac{5}{8}\cos t + \frac{5}{16}\cos 3t + \frac{1}{16}\cos 5t$

$\cos^6 t = \frac{5}{16} + \frac{15}{32}\cos 2t + \frac{3}{16}\cos 4t + \frac{1}{32}\cos 6t$

18. $\int (5\cos^3 t - 6\cos^4 t + 5\cos^5 t - 12\cos^6 t)\,dt$

these are $[x]_B$. find $[x]_C$ and integrate.

$$\underset{C \leftarrow B}{P}[x]_B = P^{-1}[x]_B = P^{-1}\begin{bmatrix} 0 \\ 0 \\ 0 \\ 5 \\ -6 \\ 5 \\ -12 \end{bmatrix} = \begin{bmatrix} -6 \\ 55/8 \\ -69/8 \\ 45/16 \\ -3 \\ 5/16 \\ -3/8 \end{bmatrix}$$

$$\int \left(-6 + \frac{55}{8}\cos t - \frac{69}{8}\cos 2t + \frac{45}{16}\cos 3t - 3\cos 4t + \frac{5}{16}\cos 5t - \frac{3}{8}\cos 6t\right)$$

$$= -6t + \frac{55}{8}\sin t - \frac{69}{16}\sin 2t + \frac{15}{16}\sin 3t - \frac{3}{4}\sin 4t + \frac{1}{16}\sin 5t - \frac{1}{16}\sin 6t + C$$

4.9 p.260 # 1,3,5,9,11,13,15

1. a) $P = \begin{matrix} \\ N \\ M \end{matrix}\begin{bmatrix} N & M \\ .7 & .6 \\ .3 & .4 \end{bmatrix}$ b) $(P-I)X=0$ $\begin{bmatrix} -.3 & .6 \\ .3 & -.6 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}$ $\bar{x}=\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $X=\begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}$

b) $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ c) $\begin{bmatrix} .7 & .6 \\ .3 & .4 \end{bmatrix}^2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} .67 \\ .33 \end{bmatrix}$ 33%

3. a) $P = \begin{matrix} \\ H \\ I \end{matrix}\begin{bmatrix} H & I \\ .95 & .45 \\ .05 & .55 \end{bmatrix}$ b) Tuesday: $P\begin{bmatrix} .8 \\ .2 \end{bmatrix} = \begin{bmatrix} .85 \\ .15 \end{bmatrix}$ 15% sick

Wed: $P\begin{bmatrix} .85 \\ .15 \end{bmatrix} = \begin{bmatrix} .875 \\ .125 \end{bmatrix}$ 12.5% sick

c) $P^2\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} .925 \\ .075 \end{bmatrix}$ .925 prob. of healthy

→ 9. $P = \begin{bmatrix} .2 & 1 \\ .8 & 0 \end{bmatrix}$. no, not all strictly positive

11. $\begin{bmatrix} .95 & .03 \\ .05 & .97 \end{bmatrix} = M$ a) $(M-I)X=0$ not ex. 1

$\begin{bmatrix} -.05 & .03 \\ .05 & -.03 \end{bmatrix} \sim \begin{bmatrix} 1 & -.6 \\ 0 & 0 \end{bmatrix}$ $X=\begin{bmatrix} 3 \\ 5 \end{bmatrix}$

a) See #1 part b* $\begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}$ b) 2/3 listening to news eventually

13. $P = \begin{bmatrix} .95 & .45 \\ .05 & .55 \end{bmatrix}$ a) $(P-I)X=0$

$\begin{bmatrix} -.05 & .45 \\ .05 & -.45 \end{bmatrix} \sim \begin{bmatrix} 1 & -9 \\ 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 9 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} .9 \\ .1 \end{bmatrix} = X$

b) .10 probability that ill after many days whether ill or not today

15

$$\begin{matrix} & C & \text{other} \\ \begin{matrix} C \\ \text{other} \end{matrix} & \begin{bmatrix} .9821 & .0029 \\ .0179 & .9971 \end{bmatrix} \end{matrix}$$

$$P - I = \begin{bmatrix} -.0179 & .0029 \\ .0179 & -.0029 \end{bmatrix} \sim \begin{bmatrix} 1 & -.162011 \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} .162011 \\ 1 \end{bmatrix}$$

Sum = 1.162011

$$X = \begin{matrix} \begin{bmatrix} .139423 \\ .860577 \end{bmatrix} & \begin{matrix} C \\ \text{other} \end{matrix} \end{matrix}$$

~13.9% in Cali